

Muhammed İkbāl Yılmaz

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EDUCATION

Ankara Hacı Bayram Veli University

Ankara, Turkey

Bachelor of Science in Econometrics

Sep 2017 – Jul 2024

- **Relevant Coursework:** Statistical Modeling, Time Series Analysis, Artificial Intelligence for Business, Stochastic Processes, Regression Diagnostics, Linear Algebra, Statistical Tests, Financial Econometrics, Sampling with Stata.

ACADEMIC RESEARCH & TÜBİTAK PROJECTS

Classification of Brand Perception Using Random Forest

IDSSC'24 (Oct 2024)

International Data Science and Statistics Congress – Oral Presentation

Python, Scikit-learn, Imbalanced-learn

- Developed a multi-class Random Forest model for brand perception; utilized SMOTE to balance multi-class distributions, improving recall and precision metrics across minority classes. Analyzed decision trees and entropy node splits.

TÜBİTAK 2209-A: Brand Preference and Loyalty Dynamics

Sep 2023 – Sep 2024

Project Principal Investigator (Yürütücü) – Supported by TÜBİTAK BİDEB

SPSS, Stata, Python

- Led end-to-end data pipeline on $N = 407$ dataset: survey design, data cleaning, Q-Q normality checks, Spearman rank correlations, Chi-square (χ^2) independence tests, and ANOVA with Tukey's HSD post-hoc tests.

Empirical Analysis on Brand Preferences

UYİK-2024 (May 2024)

V. International Applied Statistics Congress – Proceedings Paper

- Conducted two-sample t -tests and Pearson correlation matrices to quantify consumer loyalty and trust drivers.

SELECTED TECHNICAL REPOSITORIES & MODELING PROJECTS

Macro-Financial Modeling: BLUE vs. Machine Learning

GitHub: [bitcoin-macro-blueml](#)

Python (Statsmodels, Scikit-learn, Pandas, Seaborn)

- Evaluated Gauss-Markov (BLUE) assumptions against multicollinearity and heteroskedasticity; benchmarked classical Ordinary Least Squares (OLS) regression against non-linear Random Forest architectures.

Return Dynamics under Risk Factors (ASELSAN Case Study)

GitHub: [Return-Dynamics-ASELSAN](#)

Jupyter Notebook, Time Series Econometrics & ML

- Modeled equity return dynamics under sovereign risk factors using a multivariate ARX framework and predictive machine learning models; performed rigorous diagnostic and robustness checks on financial time series.

Time Series Stationarity Pipelines (ADF & ARMA)

GitHub: [ADF_Unit_Root](#) / [ARMA_CO2_gold](#)

Python (Statsmodels, Pandas, Matplotlib)

- Implemented Automated Augmented Dickey-Fuller (ADF) unit root testing workflows to identify stationarity in macroeconomic indicators; built AutoRegressive Moving Average (ARMA) models with residual error diagnostics.

Algorithmic Simulations & Statistical Computing

GitHub: [LCG-R](#) / [ANN-Classification](#)

- **LCG Pseudo-Random Generator (R):** Built modular random generator satisfying Hull–Dobell conditions for simulation.

- **Deep Learning Classifier (Python):** Designed Artificial Neural Network (ANN) pipelines for classification tasks.

LICENSES & CERTIFICATIONS

- **Uygulamalı Kaggle:** Bilgi Teknolojileri ve İletişim Kurumu (BTK Akademi)

Jul 2024

- **Foundations: Data, Data, Everywhere:** Google Coursera

Oct 2024

- **Data & AI Class | IT Class:** Türkiye İş Bankası ProSchool & Toptalent

Aug 2024

- **Regression Analysis Fundamentals:** IIBA (Endorsed Education Provider)

Jul 2024

- **English Proficiency (CEFR C1 - Advanced):** Toptalent Business School

Jul 2024

- **Stata 101:** Analiz Akademi (Udemy)

May 2024

- **Thriving as a Project Manager in the Age of AI:** PMI & LinkedIn Learning

Aug 2024

- **Cert Prep: Word Expert (MOS 2019):** LinkedIn Learning

Aug 2024

TECHNICAL SKILLS & LANGUAGES

- **Programming & Tools:** Python (pandas, numpy, scikit-learn, statsmodels), R, SQL, Stata, SPSS, EViews, LaTeX, MATLAB, Excel.

- **Methodologies:** Machine Learning (Classification, Decision Trees, ANN), Time Series (ADF, ARMA, ARX), Hypothesis Testing, Pseudo-Random Number Generation (LCG), Econometric Modeling.

- **Languages:** Turkish (Native), English (C1 - Advanced), German (B1 - Intermediate), Arabic (Elementary).

- **Volunteering:** TEMA Foundation (Volunteer).